1. APPLICATIONS OF COMPUTER GRAPHICS (15 MARK)

Computer Graphics is used to create, store, and manipulate images using computers. It plays a vital role in many real-world applications.

Major Applications

1. Computer-Aided Design (CAD)

- Used in engineering, architecture
- Designing buildings, cars, machines

Example: AutoCAD

2. Computer-Aided Manufacturing (CAM)

- Used in production systems
- Controls machines using graphics

3. Entertainment & Animation

- Movies, cartoons, games
- 3D animation and visual effects

4. Education & Training

- Simulation-based learning
- Virtual labs

5. Medical Imaging

- MRI, CT scan visualization
- Surgery simulation

6. Graphical User Interface (GUI)

- Windows, icons, menus
- User-friendly interaction

7. Simulation & Modeling

- Flight simulators
- Weather prediction

8. Image Processing

- Editing photos
- Enhancing images

9. Virtual Reality (VR)

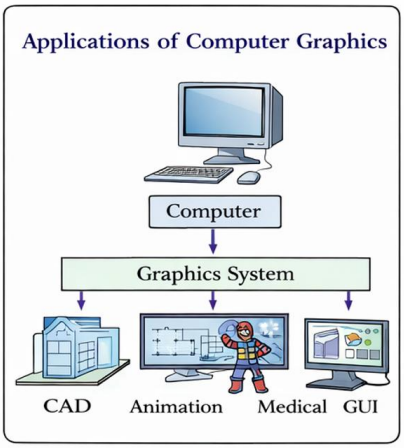
- Immersive environments

- Gaming and training

10. Scientific Visualization

- Graphs, charts, data visualization

Diagram



2. COMPOSITE TRANSFORMATION (VERY IMPORTANT)

Composite transformation is the process of combining multiple transformations into a **single transformation matrix**.

Need

- Reduce computation
- Improve efficiency
- Perform multiple operations at once

Example Transformations

- Translation
- Rotation
- Scaling

Mathematical Concept

If we apply:

- Scaling (S)
- Rotation (R)
- Translation (T)

Then composite matrix:

$$M = T \times R \times S$$

Algorithm

START

1. Identify transformations required
2. Write transformation matrices
3. Multiply matrices in correct order
4. Get composite matrix
5. Apply matrix to object

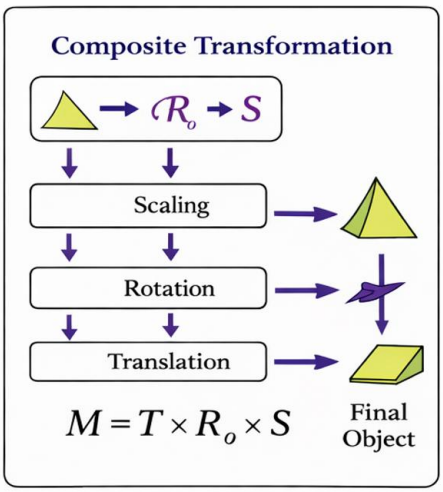
STOP

Important Note

Order matters:

- TR ≠ RT

Diagram



Advantages

- Faster computation
- Efficient
- Easy implementation

3. SWEEP REPRESENTATION & CSG (WITH FIGURES)

(A) SWEEP REPRESENTATION

Definition

A 3D object is created by moving a 2D shape along a path.

Types

- Translational Sweep

- Rotational Sweep

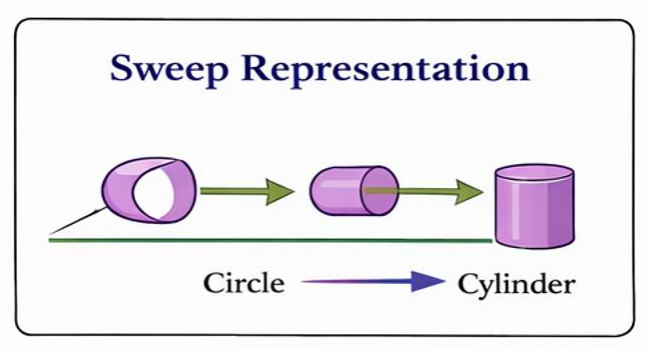
Algorithm (START–STOP)

START

1. Define 2D shape
2. Define path
3. Move shape along path
4. Generate surface
5. Form 3D object

STOP

Diagram



Advantages

- Easy modeling
- Useful in engineering

(B) CONSTRUCTIVE SOLID GEOMETRY (CSG)

Definition

Objects are created using **Boolean operations** on primitives.

Operations

- Union (∪)
- Intersection (∩)
- Difference (−)

Algorithm (START–STOP)

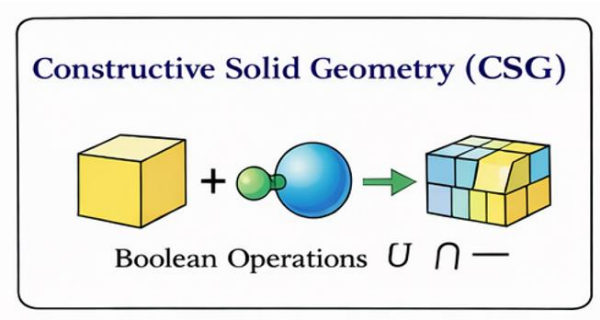
START

1. Define primitives (cube, sphere)
2. Select operation
3. Apply Boolean operation

4. Generate final object

STOP

Diagram



Advantages

- Accurate modeling
- Easy editing

Conclusion

Sweep and CSG are powerful methods for creating complex 3D objects.

4. STEPS IN DESIGN OF ANIMATION SEQUENCE (VERY IMPORTANT)

Animation is the process of creating motion by displaying a sequence of images.

Steps in Animation Design

1. Storyboard Layout

- Plan animation sequence
- Define scenes

2. Object Definition

- Define shapes and models

3. Key Frame Specification

- Define important frames
- Start and end positions

4. In-between Frames (Tweening)

- Generate intermediate frames
- Smooth motion

5. Motion Specification

- Define movement path
- Speed and direction

6. Rendering

- Generate final images

7. Display

- Show frames in sequence

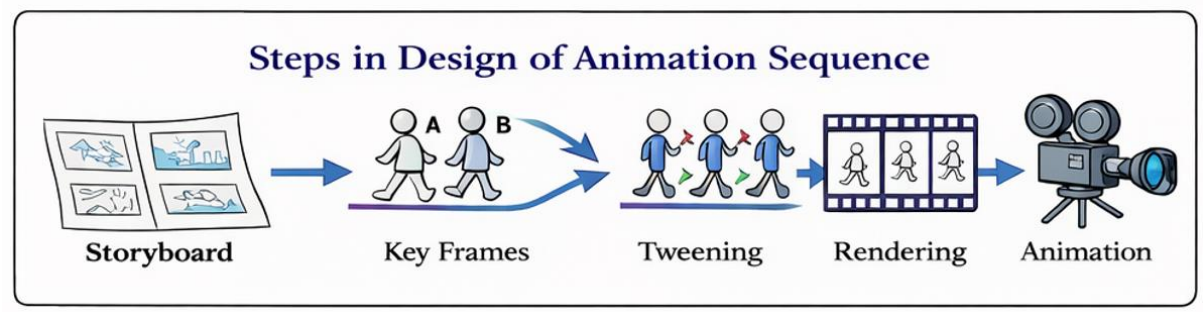
Algorithm

START

1. Create storyboard
2. Define objects
3. Specify key frames
4. Generate intermediate frames
5. Apply motion
6. Render frames
7. Display animation

STOP

Diagram



Applications

- Movies
- Games
- Simulation